BUSINESS REPORT ON AUTOMOBILES AND CREDIT CARD

Vasanthi Rao

Contents: - Problem Statement……
Solution…

A. What is the important technical information about the dataset that a database administrator would be interested in? (Hint: Information about the size of the dataset and the nature of the variables)

B. Take a critical look at the data and do a preliminary analysis of the variables. Do a quality check of the data so that the variables are consistent. Are there any discrepancies present in the data?.....

C. Explore all the features of the data separately by using appropriate visualization and draw insights that can be utilized by the business. Univariate Analysis for Numerical Variables Univariate Analysis for Categorical Variables.

D. Understanding the relationships among the variables in the dataset is crucial for every analytical project. Perform analysis on the data fields to gain deeper insights. Comment on your understanding of the data..Bivariate Analysis/Relationship among the Variables……Correlation between the Variables..

E. Employees working on the existing marketing campaign have made the following remarks. Based on the data and your analysis state whether you agree or disagree with their observations. Justify your answer Based on the data available..... a. Steve Roger says "Men prefer SUV by a large margin, compared to the women" b. Ned Stark believes that a salaried person is more likely to buy a Sedan. c. Sheldon Cooper does not believe any of them; he claims that a salaried male is an easier target for an SUV sale over a Sedan Sale.

F. From the given data, comment on the amount spent on purchasing automobiles across the following categories. Comment on how a Business can utilize the results from this exercise. Give justification along with presenting metrics/charts used for arriving at the conclusions. Give justification along with presenting metrics/charts used for arriving at the conclusions

F1) Gender..... F2) Personal Loan.....

G. From the current data set comment if having a working partner leads to the purchase of a higher-priced car.

H. The main objective of this analysis is to devise an improved marketing strategy to send targeted information to different groups of potential buyers present in the data. For the

current analysis use the gender and Marital status – fields to arrive at groups with similar purchase history.

Problem 1

Analysts are required to explore data and reflect on the insights. Clear writing skill is an integral part of a good report. Note that the explanations must be such that readers with minimum knowledge of analytics is able to grasp the insight.

Austo Motor Company is a leading car manufacturer specializing in SUV, Sedan, and Hatchback models. In its recent board meeting, concerns were raised by the members on the efficiency of the marketing campaign currently being used.The board decides to rope in an analytics professional to improve the existing campaign.

1.You as an analyst have been tasked with performing a thorough analysis of the data and coming up with insights to improve the marketing campaign.

Q.A. What is the important technical information about the dataset that a database administrator would be interested in?

Answer : A database administrator would be interested in the size of the dataset and nature of the variables in the dataset.

The size of the dataset can be identifed as the size of the rows and columns of the dataset as how many number of rows and columns are present in the dataset.Similarly the nature of the variables can be identifed as each row or column has a individual nature such as object,numeric etc.

Nature of the variables

```
<class
'pandas.core.frame.DataFrame'>
RangeIndex: 1581 entries, 0 to
1580
Data columns (total 14 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
```

```
 0   Age                 1581 non-null   int64
 1   Gender              1528 non-null   object
 2   Profession          1581 non-null   object
 3   Marital_status      1581 non-null   object
 4   Education           1581 non-null   object
 5   No_of_Dependents    1581 non-null   int64
 6   Personal_loan       1581 non-null   object
 7   House_loan          1581 non-null   object
 8   Partner_working     1581 non-null   object
 9   Salary              1581 non-null   int64
 10  Partner_salary      1475 non-null   float64
 11  Total_salary        1581 non-null   int64
 12  Price               1581 non-null   int64
 13  Make                1581 non-null   object dtypes: float64(1), int64(5),
object(8)
memory usage: 173.1+ KB
```

type pandas.core.frame.DataFrame
size 22134
no. of rows:1581
no. of columns:14

Q.B. Take a critical look at the data and do a preliminary analysis of the variables. Do a quality check of the data so that the variables are consistent. Are there any discrepancies present in the data?

Preliminary analysis of the data

Step 1 : Dataset analysis
First of all, let's see what we have in the dataset.

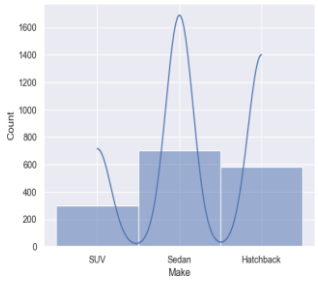
```
List    ['Age',
         'Gender',
         'Profession',
         'Marital
         _status',
         'Educati
         on',
         'No_of_De
         pendents',
         'Personal_l
```

oan',
'House_loan',
'Partner_
working',
'Salary',
'Partner
_salary',
'Total_s
alary',
'Price',
'Make']

Step 2 : Analysis questions

To perform a preliminary analysis, we should outline a set of questions that we want to answer as a result of our analysis. 1.what is the highest making vehicle? 2.Which vehicle makes the most price? 3.Is there any relationship between marital_status and no of dependants? 4.Which gender has the highest profession at the certain age count?

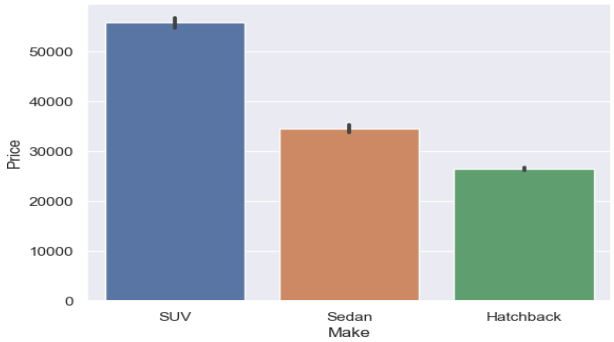
1.what is the highest making vehicle?



Observation :

From the above graph we can say that the Sedan vehicle is the highest.

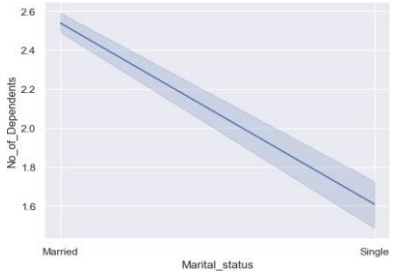
2.Which vehicle makes the most price?



Observation:

Suv vehicle makes the highest price above the range of 50,000.

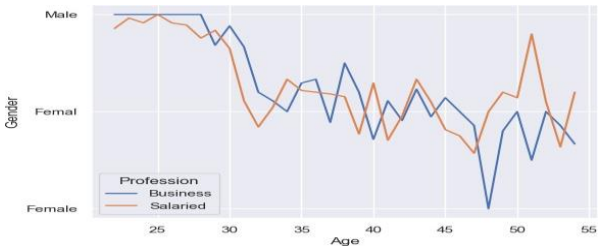
3.Is there any relationship between marital_status and no of dependants?



Observation :

Yes,there is a relationship between the Marital_status and No_of_Dependents. The no of dependents are more for the married persons than the single persons. It has a declining rate from married to single.

4.Which gender has the highest profession at the certain age count?



Observation :

The male has highest both Business and salaried professions at the below age of 25

Data quality check :

Step 1 :
printing top 5 rows

	Age	Gender	Profession	Marital_status	Education	No_of_Dependents	Personal_loan	Ho
0	53	Male	Business	Married	Post No Graduate	4		
1	53	Femal	Salaried	Married	Post Graduate	4		Yes
2	53	Female	Salaried	Married	Post No Graduate	3		
3	53	Female	Salaried	Married	Graduate	2		Yes
4	53	Male	Salaried	Married	Post No Graduate	3		

Shape of the dataset

(1581, 14)

Checking Index

array([0,1,2, ..., 1578, 1579, 1580], dtype=int64)

<class
'pandas.core.frame.DataFrame'>

RangeIndex: 1581 entries, 0 to 1580
Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	Age	1581 non-null	int64
1	Gender	1528 non-null	object
2	Profession	1581 non-null	object
3	Marital_status	1581 non-null	object
4	Education	1581 non-null	object
5	No_of_Dependents	1581 non-null	int64
6	Personal_loan	1581 non-null	object
7	House_loan	1581 non-null	object
8	Partner_working	1581 non-null	object
9	Salary	1581 non-null	int64
10	Partner_salary	1475 non-null	float64
11	Total_salary	1581 non-null	int64
12	Price	1581 non-null	int64
13	Make	1581 non-null	object

dtypes: float64(1), int64(5), object(8)
memory usage: 173.1+ KB

array(['Male', 'Femal', 'Female', nan, 'Femle'], dtype=object)

Age False
Gender True
Profession False
Marital_status False
Education False
No_of_Dependents False
Personal_loan False
House_loan False
Partner_working False
Salary False
Partner_salary True
Total_salary False
Price False
Make False
dtype: bool

'Partner_salary'

```
array([70700., 70300., 60700., 60200., 60500., 50800., 40400.,     0.,
       70600., 60600., 60300., 45500., 40200., 40700.,   600.,   700.,
       27800., 70000., 40100., 40900., 27000.,    nan, 40500., 50400.,
       80400., 60900., 60100., 70200., 30000., 40300., 40800., 70800.,
         900., 45000., 40600., 50700., 80500., 27700., 35800., 26600.,
       60000., 27900., 60800., 40000., 70100., 38100., 38200., 30200.,
       38500., 50900., 35100., 38700., 38300., 38900., 23200., 24700.,
       30800., 28100., 38400., 38000., 23100., 30100., 25700., 70400.,
         200., 30900., 26100., 45700.,   400., 28200., 70900., 38800.,
       45200., 26800., 60400., 30300., 25800.,   800., 38600., 24500.,
       30700., 28500., 24900., 70500., 35900., 30500., 28900., 27200.,
       45900., 25300., 35600., 25000., 25200., 26300., 35700., 23800.,
       32700., 30600., 45600., 28000., 32600., 28600., 25100., 26700.,
       24200., 25900., 22100., 25400., 28400., 28300., 25500., 24000.,
       32400., 28800., 32300., 32900., 27600., 35500., 23500., 45400.,
       28700., 32800., 23700., 32500., 35300., 45800., 27300., 22900.,
       29800., 30400., 35400., 24300., 29200., 24600.,   100., 50300.,
       32000., 32200., 24400., 32100., 22600., 26900.,   300., 20000.,
       26200., 22300., 25600.,   500., 35200., 35000.])
```

740	29	Male	Business	Married	Post No Graduate	2	
741	29	Male	Business	Married	Post Graduate	3	Yes
742	29	Male	Business	Married	Post Graduate	2	Yes
743	29	Male	Salaried	Married	Post	3	Yes
744	29	Male	Salaried	Married	Post Graduate	4	No
...	...	...	...	...	...	...	...
1576	22	Male	Salaried	Single	Graduate	2	No
1577	22	Male	Business	Married	Graduate	4	No
1578	22	Male	Business	Single	Graduate	2	No
1579	22	Male	Business	Married	Graduate	3	Yes
1580	22	Male	Salaried	Married	Graduate	4	No

Age	0
Gender	53
Profession	0
Marital_status	0
Education	0
No_of_Dependents	0
Personal_loan	0
House_loan	0
Partner_working	0
Salary	0
Partner_salary	106
Total_salary	0
Price	0
Make	0

dtype: int64

Age	0
Gender	53
Profession	0
Marital_status	0
Education	0
No_of_Dependents	0
Personal_loan	0
House_loan	0
Partner_working	0
Salary	0
Partner_salary	106
Total_salary	0
Price	0
Make	0

dtype: int64

Age	0
Gender	0
Profession	0
Marital_status	0
Education	0
No_of_Dependents	0
Personal_loan	0
House_loan	0
Partner_working	0
Salary	0
Partner_salary	0
Total_salary	0
Price	0

Make 0
dtype: int64

Number of duplicate rows = 0

Q.C Explore all the features of the data separately by using appropriate visualizations and draw insights that can be utilized by the business.

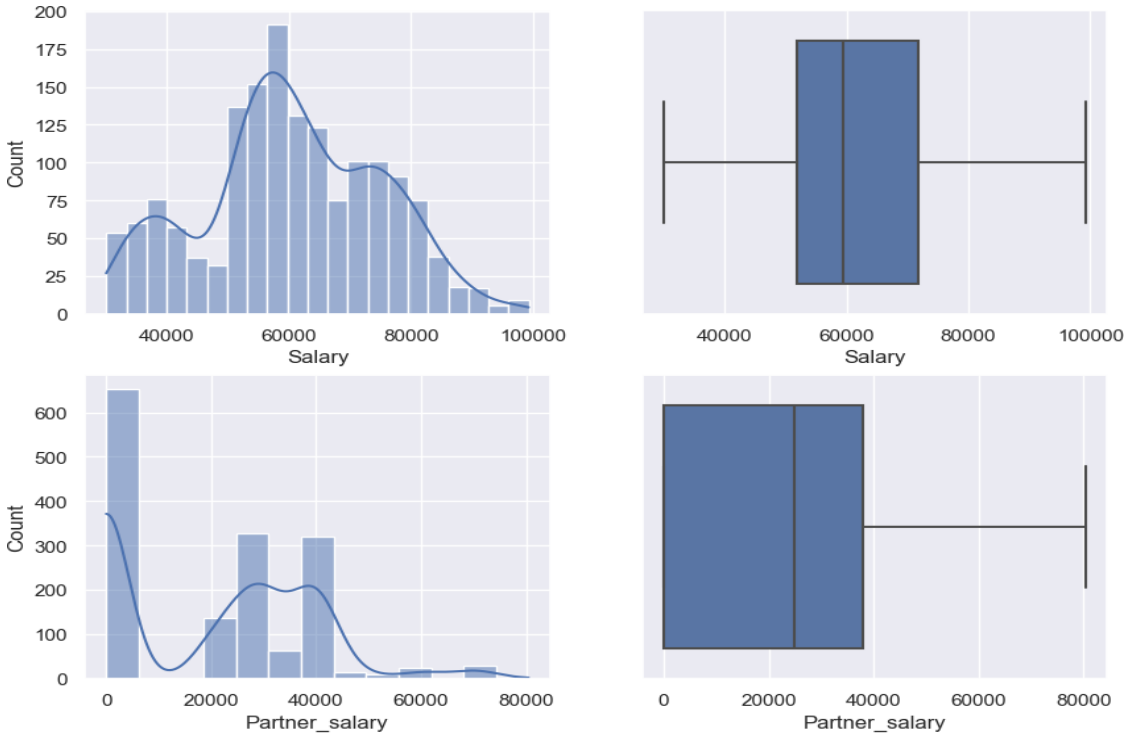
Univariate Analysis

	Salary	Partner_salary
count	1581.000000	1581.000000
mean	60392.220114	20225.559322
std	14674.825044	18905.183912
min	30000.000000	0.000000
25%	51900.000000	0.000000
50%	59500.000000	24900.000000
75%	71800.000000	38000.000000
max	99300.000000	80500.000000

Insights

The Salary of working personranges from minimum of 30k to maximum99k.

Mean of salary is higher than partnersalary which meansthe total salaryhas the majority of Salary rather than Partner Salary.



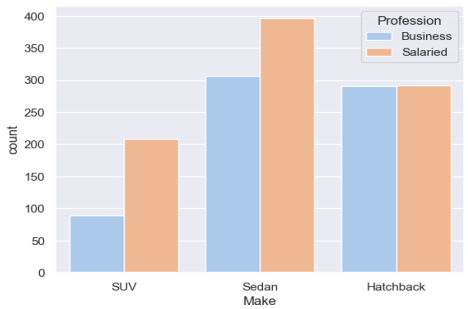
Insights:

Both Salary and Partner_salary does not have any outliers.

Categorical Variable

Sedan	0.444023
Hatchback	0.368121
SUV	0.187856

Name: Make, dtype: float64



Insights:

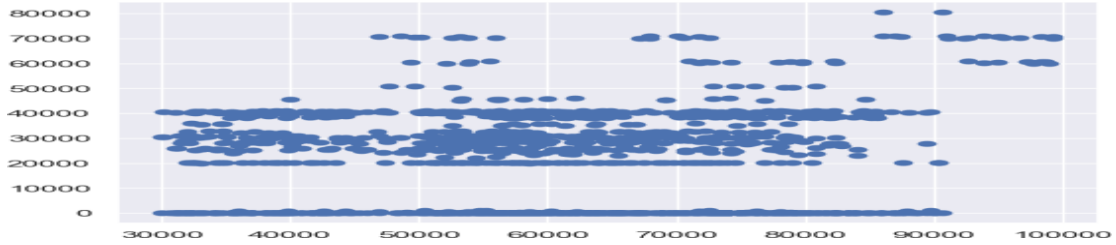
1.Sedan has the highest making count.

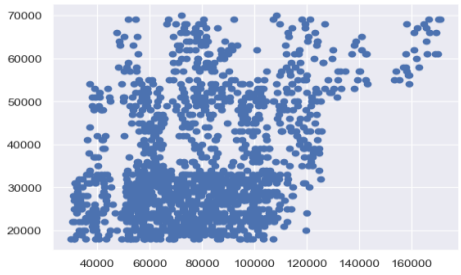
2.Salaried people has the highest cars than Business people for both SUV and Sedan vehicles.

3.Both Business and Salaried persons have equal Hatchback vehicles.

Bivariate Analysis

Numeric vs Numeric



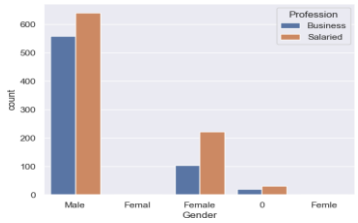


Insights:

salary and partner salary does not have a pattern of increasing or decreasing which results in non identification of relationship Total salary and price has a pattern of increasing range from 40k to 140k which results in mild identification of a positive relationship.

Categorical vs Categorical

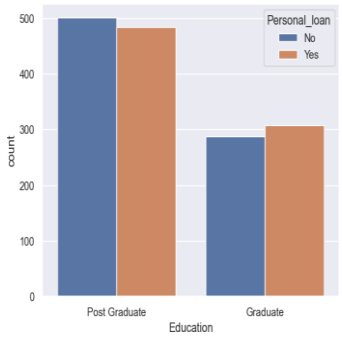
1.For Gender and profession



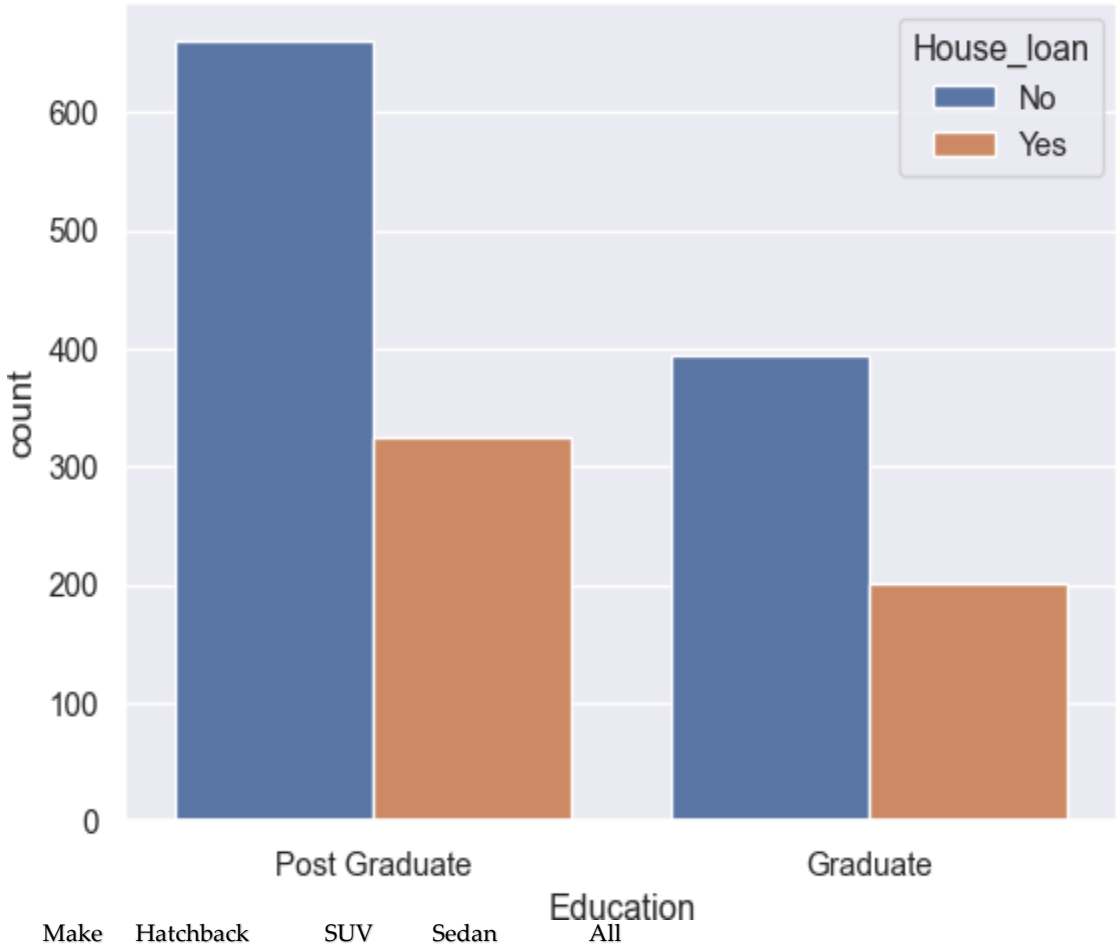
Profession	Business	Salaried	All
Gender			
0	0.013283	0.020240	0.033523
Femal	0.000000	0.000633	0.000633
Female	0.065781	0.141050	0.206831
Femle	0.000633	0.000000	0.000633
Male	0.353574	0.404807	0.758381
All	0.433270	0.566730	1.000000

Insights :

Here from the above plot the male salaried person has the highest rate of count.
For the Education and Personal_loan.

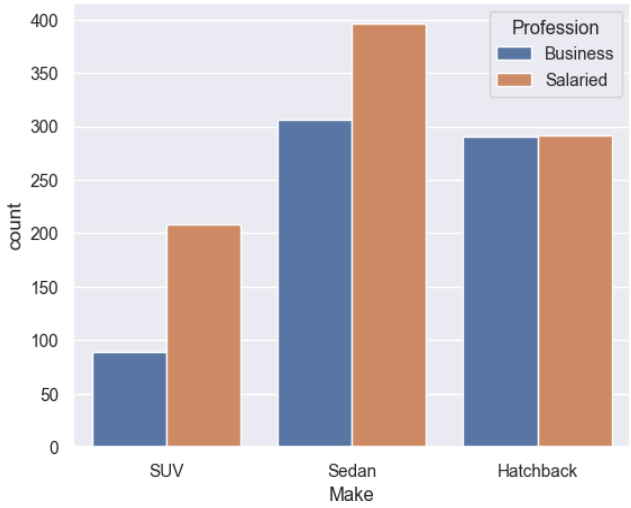


Personal_loan	No	Yes	All
Education			
Graduate	0.182163	0.194813	0.376977
Post Graduate	0.316888	0.306135	0.623023
All	0.499051	0.500949	1.000000



Make	Hatchback	SUV	Sedan	All
Gender				
0	0.001265	0.003795	0.028463	0.033523
Femal	0.000000	0.000633	0.000000	0.000633
Female	0.009488	0.108159	0.089184	0.206831

Femle	0.000000	0.000633	0.000000	0.000633
Male	0.357369	0.074636	0.326376	0.758381
All	0.368121	0.187856	0.444023	1.000000



Insights:

Salaried people has the highest Sedan vehicles and the Business people has the lowest suv vehicles.

similarly Business people has the highest Sedan vehicles and the Salaried people has the lowest suv vehicles.

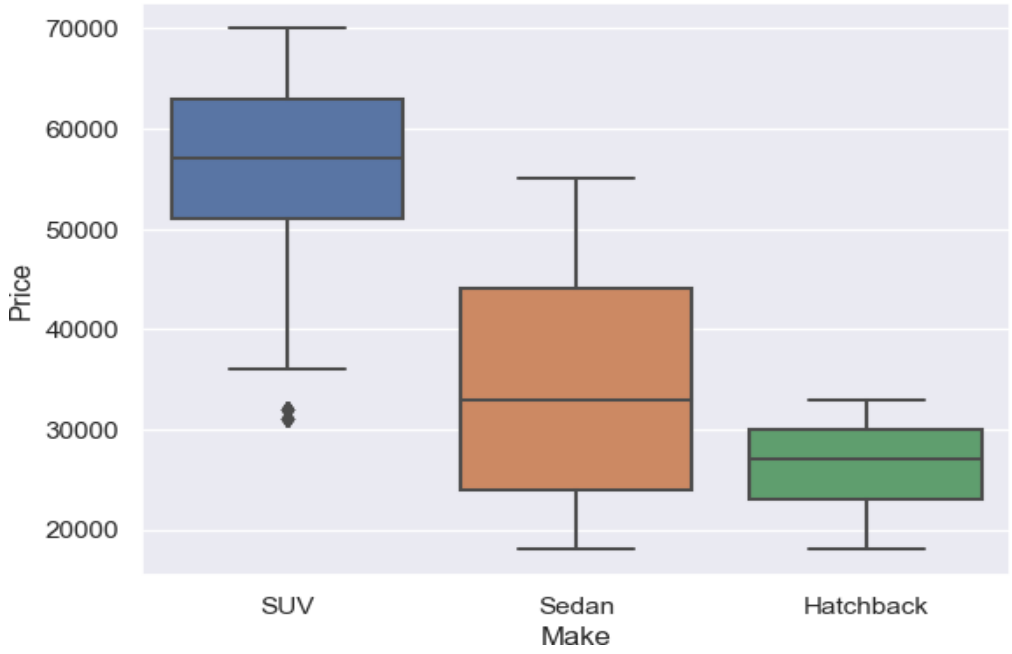
Insights:

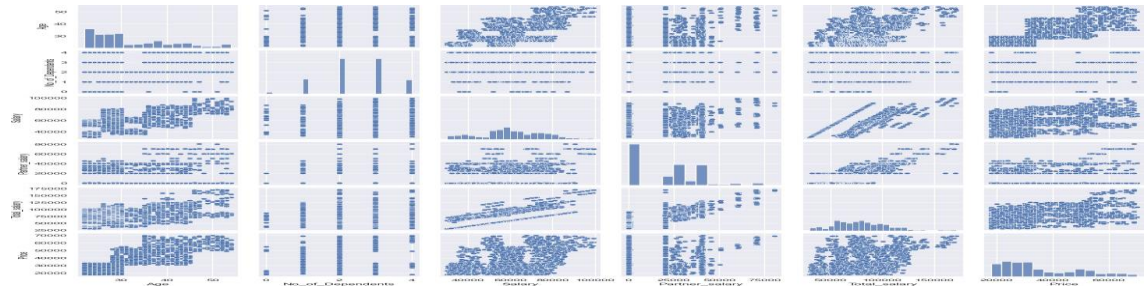
Suv has an outlier

Hatchback has the low boxplot range

Categorical vs Numeric



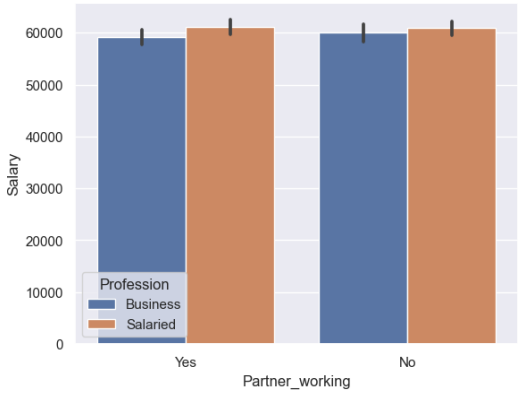




Insights:

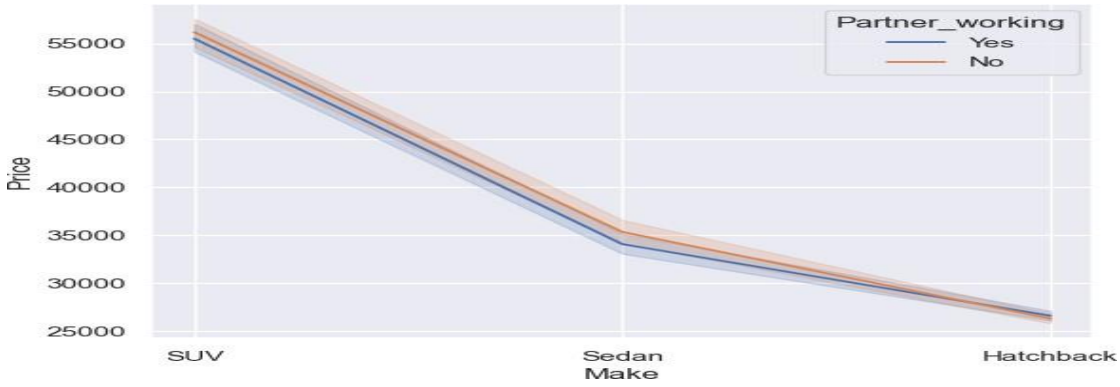
The price has correlated with the Age

Q.D. Understanding the relationships among the variables in the dataset is crucial for every analytical project. Perform analysis on the data fields to gain deeper insights. Comment on your understanding of the data.



Insights:

The people who has the partners has the highest salary in the salaried profession.

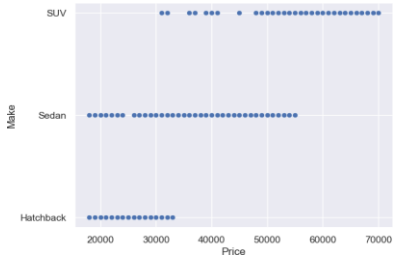


Insights:

The people who has the partners has the highest price in the suv vehicles to buy.

The price is decreased from SUV vehicles to the Hatchback vehicles.

The Hatch back has the lowest price when there are no partners who are working.



Insights:

The SUV vehicle has the highest price of points.

The hatchback has the lowest price of points.

Age	Gender	Marital_status	Education	No_of_Dependents	Personal_loan	House_loan	Partner_working	Salary	Partner_salary	Total_salary	Price	Make	
0	53	Male	Married	Post Graduate	4	No	No	Yes	99300	70700.0	1700000	61000	SUV

Age	Gender	Marital_status	Education	No_of_Dependents	Personal_loan	House_loan	Partner_working	Salary	Partner_salary	Total_salary	Price	Make	
1	53	Femal	Married	Post Graduate	4	Yes	No	Yes	95500	70300.0	1658000	61000	SUV
2	53	Female	Married	Post Graduate	3	No	No	Yes	97300	60700.0	1580000	57000	SUV
3	53	Female	Married	Graduate	2	Yes	No	Yes	72500	70300.0	1428000	61000	SUV
4	53	Male	Married	Post Graduate	3	No	No	Yes	79700	60200.0	1399000	57000	SUV

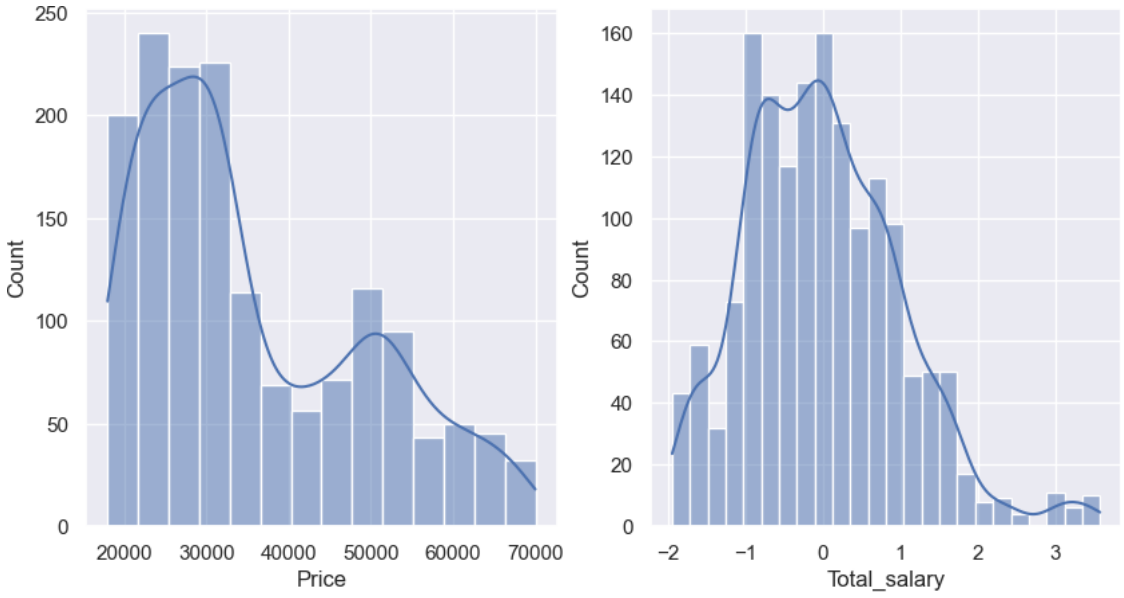
['Gender', 'Marital_status', 'Education', 'Personal_loan', 'House_loan', 'Partner_working', 'Make']
['Age', 'No_of_Dependents', 'Salary', 'Partner_salary', 'Total_salary', 'Price']

	Age	No_of_Depen	Salary	Partner Salary	Total Salary	Price
0	2.502317	1.634953	2.652167	2.670718	3.538836	1.863796
1	2.502317	1.634953	2.393138	2.649553	3.374374	1.863796
2	2.502317	0.574715	2.515836	2.141595	3.068944	1.570311
3	2.502317	-0.485523	0.825333	2.649553	2.473747	1.863796
4	2.502317	0.574715	1.316124	2.115139	2.360190	1.570311

	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	
count	1581.000000	1581.000000	1581.000000	1581.000000	1581.000000	1581.0
mean	31.922201	2.457938	60392.220114	20225.559322	79625.996205	35597.7
std	8.425978	0.943483	14674.825044	18905.183912	25545.857768	13633.6
min	22.000000	0.000000	30000.000000	0.000000	30000.000000	18000.0

25%	25.000000	2.000000	51900.000000	0.000000	60500.000000	25000.0
50%	29.000000	2.000000	59500.000000	24900.000000	78000.000000	31000.0
75%	38.000000	3.000000	71800.000000	38000.000000	95900.000000	47000.0
max	54.000000	4.000000	99300.000000	80500.000000	171000.000000	70000.0

	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	
count	1581.000000	1.581000e+03	1.581000e+03	1.581000e+03	1.581000e+03	1.5810
mean	0.000000	-5.393114e-17	-2.157245e-16	1.438164e-16	-3.595409e-17	-1.4381
std	1.000316	1.000316e+00	1.000316e+00	1.000316e+00	1.000316e+00	1.0003
min	-1.177945	-2.605999e+00	-2.071700e+00	-1.070180e+00	-1.943239e+00	-1.2911
25%	-0.821791	-4.855233e-01	-5.788762e-01	-1.070180e+00	-7.489296e-01	-7.7756
50%	-0.346918	-4.855233e-01	-6.081861e-02	2.473353e-01	-6.367023e-02	-3.3734
75%	0.721545	5.747147e-01	7.776167e-01	9.404862e-01	6.372522e-01	8.3659
max	2.621035	1.634953e+00	2.652167e+00	3.189258e+00	3.577994e+00	2.5241

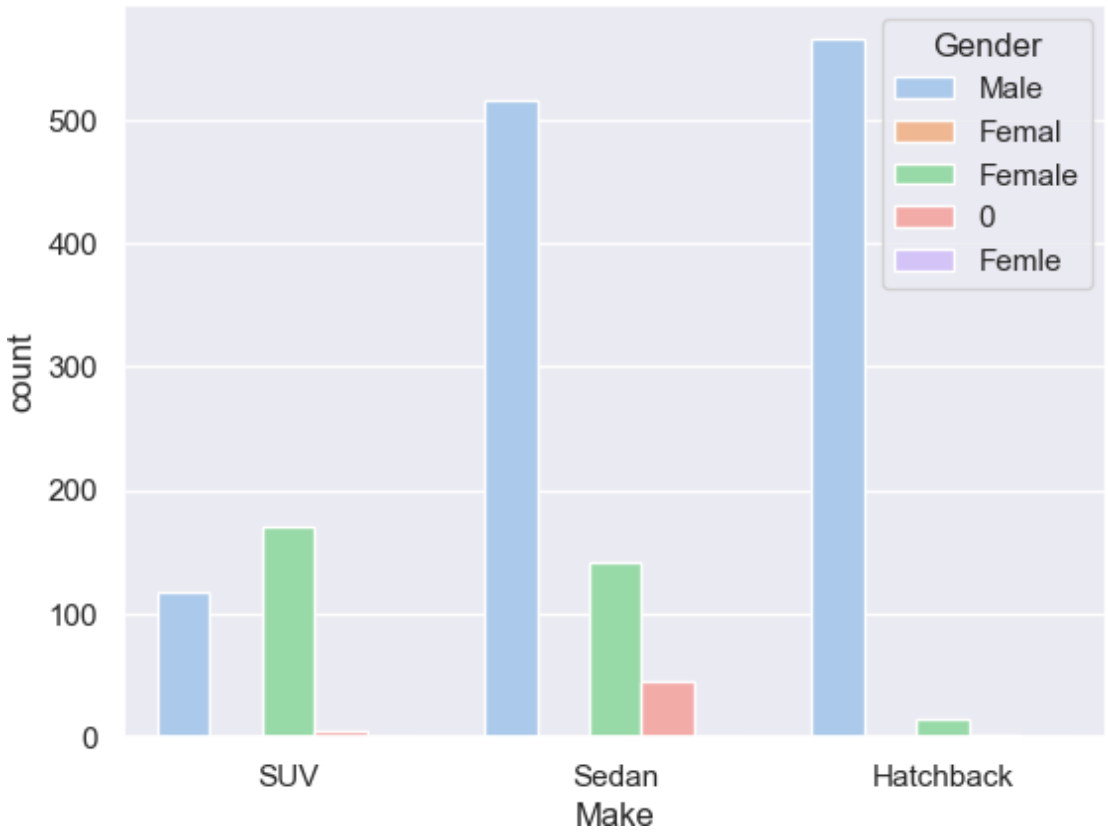


	Age	No_of_Dependents	Salary	Partner_salary	Total_salary	
count	1581.000000	1.581000e+03	1.581000e+03	1.581000e+03	1.581000e+03	1.5810
mean	0.000000	-5.393114e-17	-2.157245e-16	1.438164e-16	-3.595409e-17	-1.4381
std	1.000316	1.000316e+00	1.000316e+00	1.000316e+00	1.000316e+00	1.0003
min	-1.177945	-2.605999e+00	-2.071700e+00	-1.070180e+00	-1.943239e+00	-1.2911
25%	-0.821791	-4.855233e-01	-5.788762e-01	-1.070180e+00	-7.489296e-01	-7.7756

50%	-0.346918	-4.855233e-01	-6.081861e-02	2.473353e-01	-6.367023e-02	-3.3734
75%	0.721545	5.747147e-01	7.776167e-01	9.404862e-01	6.372522e-01	8.3659
max	2.621035	1.634953e+00	2.652167e+00	3.189258e+00	3.577994e+00	2.5241

Q.E. Employees working on the existing marketing campaign have made the following remarks. Based on the data and your analysis state whether you agree or disagree with their observations. Justify your answer Based on the data available.

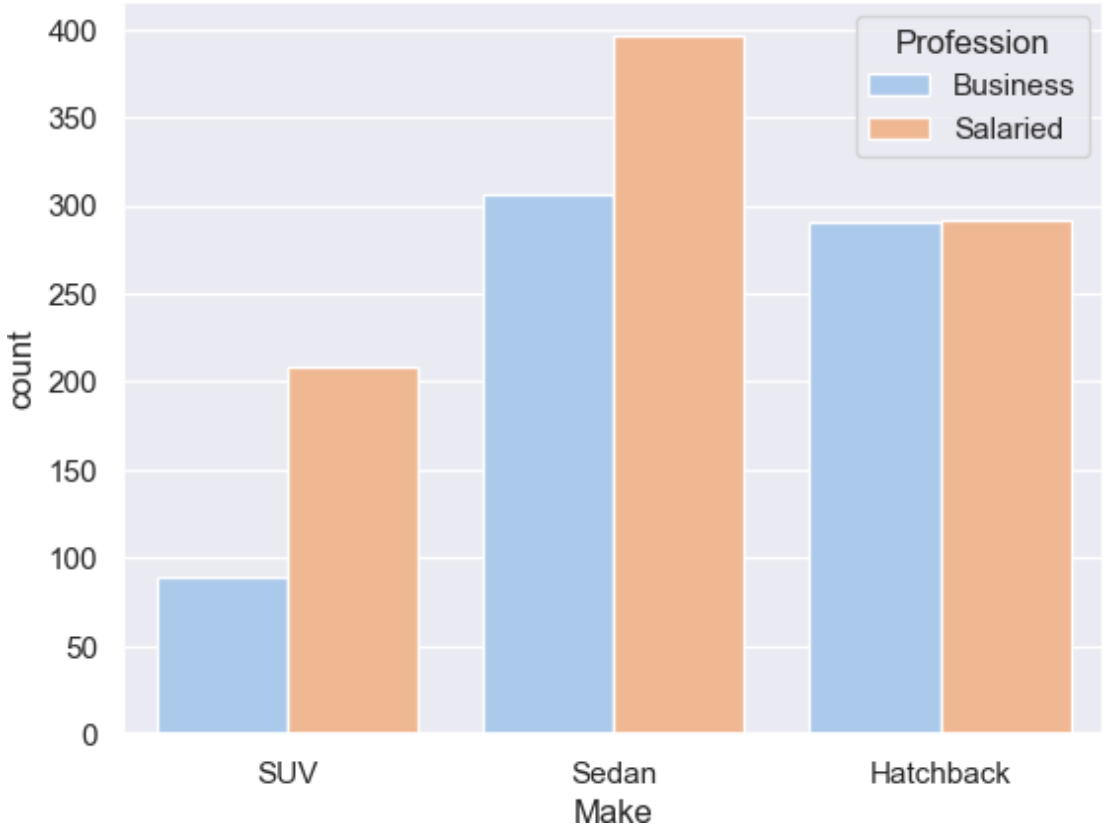
E1) Steve Roger says "Men prefer SUV by a large margin, compared to the women.



Answer:

I don't agree with Steve Rogers because as we can see from the above plot the blue margin indicates male which refers to the men. The blue margin is highest in the hatchback vehicle category, which means "Men prefers Hatchback by a large margin ,compared to the women.

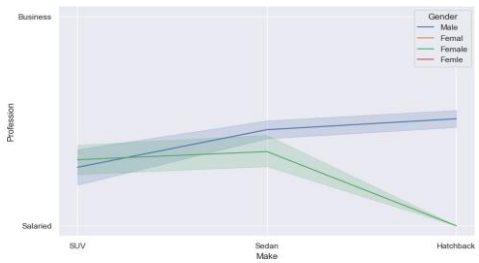
E2) Ned Stark believes that a salaried person is more likely to buy a Sedan.



Answer :

I agree with Ned Stark because as we can we see from the above plot the pink margin Indicates salaried. The pink margin is highest in the sedan vehicle category, which means " that a salaried person is more likely to buy a Sedan".

E3) Sheldon Cooper does not believe any of them; he claims that a salaried male is an easier target for a SUV sale over a Sedan Sale.

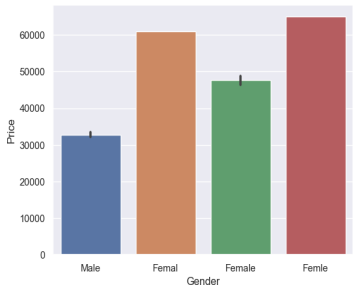


Answer :

I partially agree with Sheldon Cooper because as we can we see from the above plot the blue line Indicates Male. The blue line is nearest to the SUV vehicle category, which means " that a salaried male maybe is an easier target for a SUV sale over a Sedan Sale."

Q.F. From the given data, comment on the amount spent on purchasing automobiles across the following categories. Comment on how a Business can utilize the results from this exercise. Give justification along with presenting metrics/charts used for arriving at the conclusions.

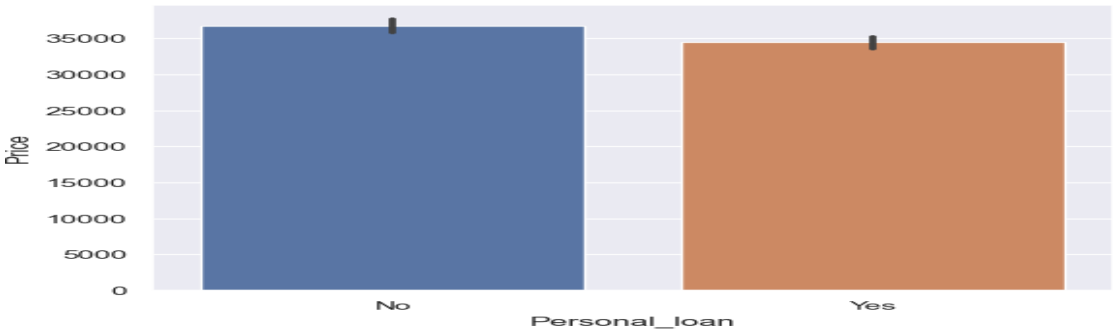
Gender



Answer:

Based on the above plot we can consider that the male has bought nearly over 30000 price and also female population has bought nearly above the 50000 price.But the highest price used for buying automobiles is the population of Femle category.Business can utilize as the interest of modifications and style of the product depends on the increasing of buyers will happen as the femle category is maybe interest in above categories of modification or style and the price rate of the automobile which results in the highest price of femle.

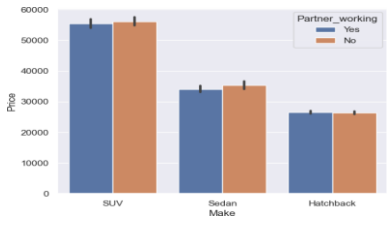
F2) Personal_loan

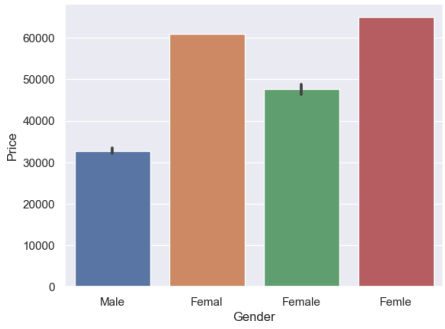


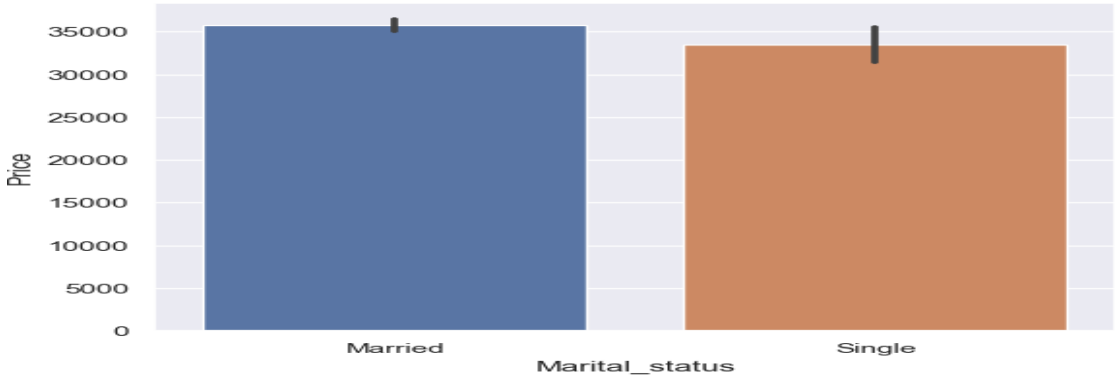
Answer:

Based on the above plot we can consider that people has personal loans which are in the range of 0 to 35000 and more people does not have personal loan that they have to pay above the 35000 price .If the people has the interest rate for having personal loan with minimum price of automobiles then the percentage of people taking personal loans for the price of automobiles to pay will be gradually increased.

Q.G. From the current data set comment if having a working partner leads to purchase of a higher priced car.







Insights: The male who are single has the minimum price of 35k.

Problem 2

Analyze the dataset and list down the top 5 important variables, along with the business justifications.

Data Shape =(8448, 28)

Null Values=

userid	0
card_no	0
card_bin_no	0
Issuer	0
card_type	0
card_source_date	0
high_networth	0
active_30	0
active_60	0
active_90	0

cc_active30	0
cc_active60	0
cc_active90	0
hotlist_flag	0
widget_products	0
engagement_products	0
annual_income_at_source	0
other_bank_cc_holding	0
bank_vintage	0
T+1_month_activity	0
T+2_month_activity	0
T+3_month_activity	0
T+6_month_activity	0
T+12_month_activity	0
Transactor_revolver	38
avg_spends_l3m	0
Occupation_at_source	0
cc_limit	0

dtype: int64

Filling the null values

userid	0
card_no	0
card_bin_no	0
Issuer	0
card_type	0
card_source_date	0
high_networth	0
active_30	0
active_60	0
active_90	0
cc_active30	0
cc_active60	0
cc_active90	0
hotlist_flag	0
widget_products	0
engagement_products	0
annual_income_at_source	0
other_bank_cc_holding	0
bank_vintage	0
T+1_month_activity	0
T+2_month_activity	0
T+3_month_activity	0
T+6_month_activity	0
T+12_month_activity	0

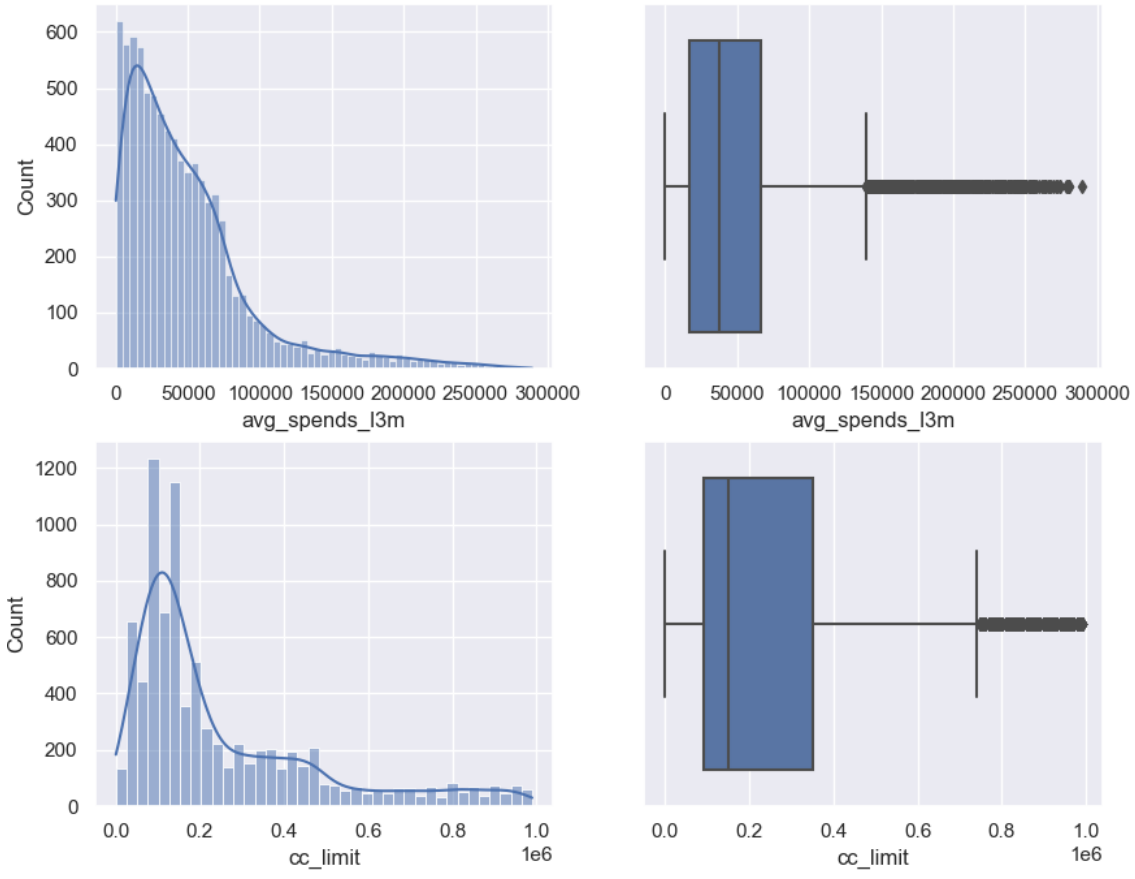
```
Transactor_revolver       0
avg_spends_l3m            0
Occupation_at_source      0
cc_limit                  0
dtype: int64

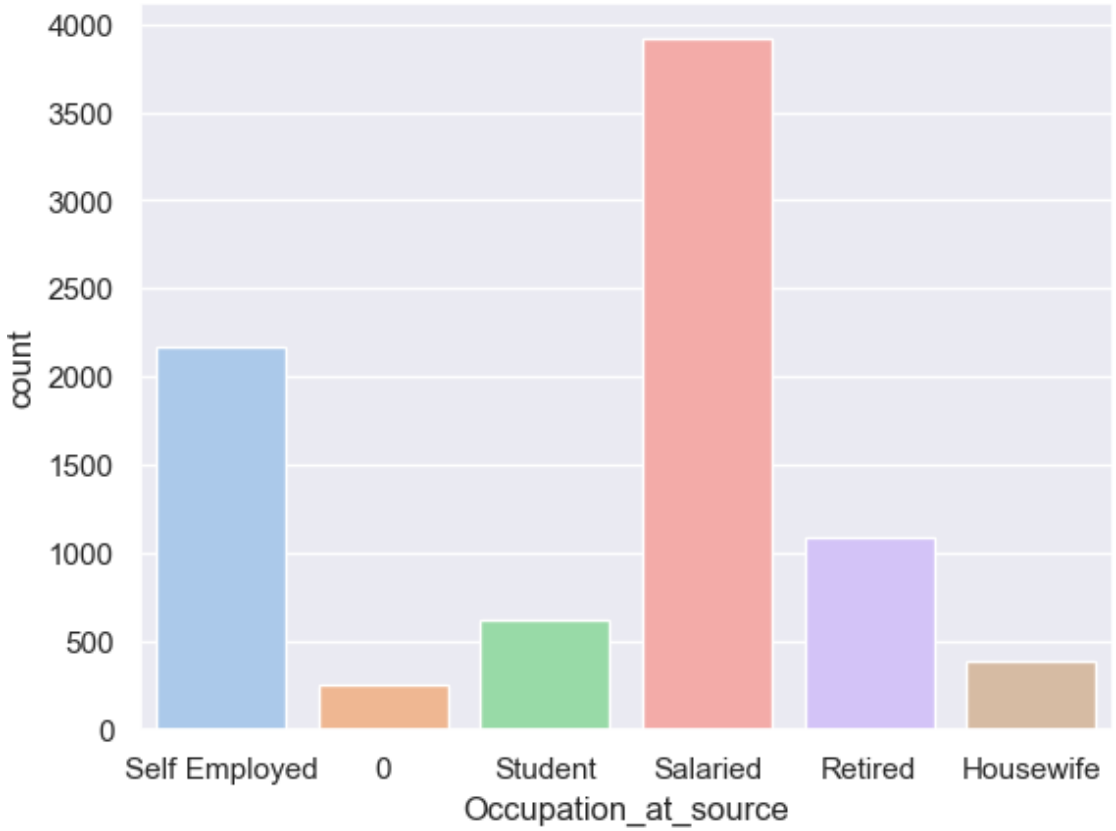


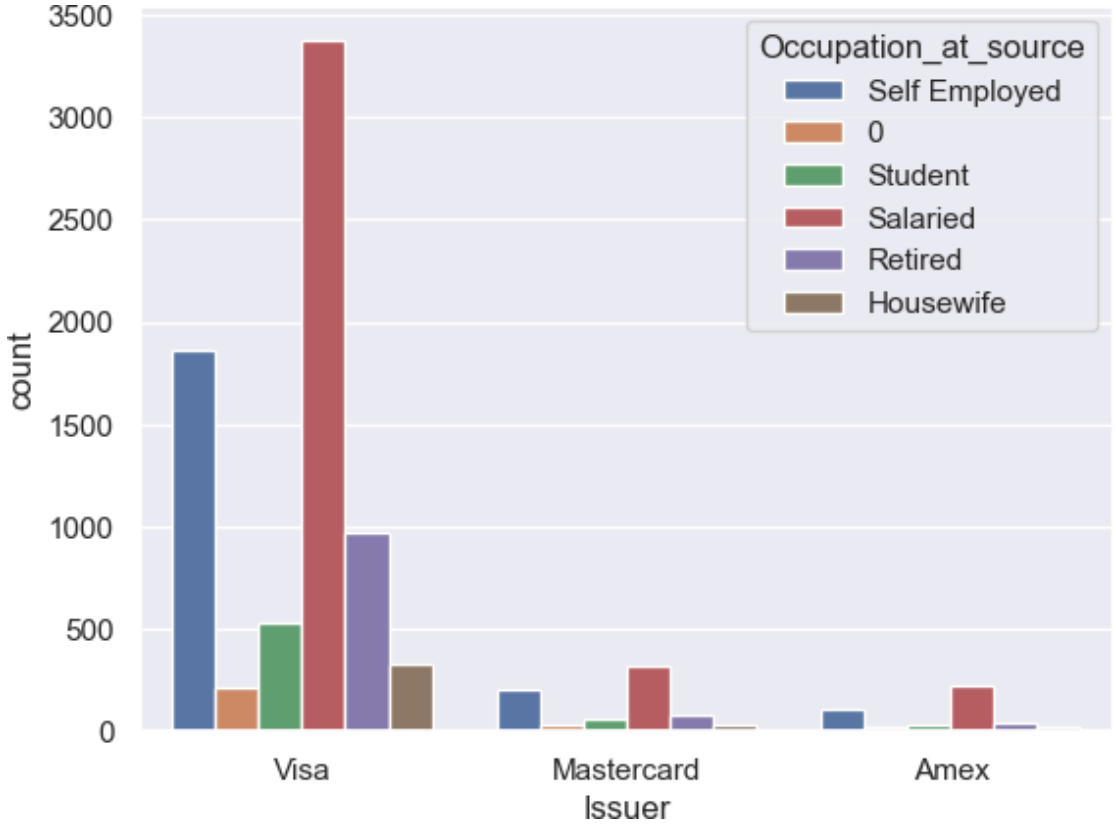
<class
'pandas.core.frame.DataFrame'>
RangeIndex: 8448 entries, 0 to
8447
Data columns (total 28 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   userid                  8448 non-null   int64
 1   card_no                 8448 non-null   object
 2   card_bin_no             8448 non-null   int64
 3   Issuer                  8448 non-null   object
 4   card_type               8448 non-null   object
 5   card_source_date        8448 non-null   datetime64[ns]
 6   high_networth           8448 non-null   object
 7   active_30               8448 non-null   int64
 8   active_60               8448 non-null   int64
 9   active_90               8448 non-null   int64
 10  cc_active30             8448 non-null   int64
 11  cc_active60             8448 non-null   int64
 12  cc_active90             8448 non-null   int64
 13  hotlist_flag            8448 non-null   object
 14  widget_products         8448 non-null   int64
 15  engagement_products     8448 non-null   int64
 16  annual_income_at_source 8448 non-null   int64
 17  other_bank_cc_holding   8448 non-null   object
 18  bank_vintage            8448 non-null   int64
 19  T+1_month_activity      8448 non-null   int64
 20  T+2_month_activity      8448 non-null   int64
 21  T+3_month_activity      8448 non-null   int64
 22  T+6_month_activity      8448 non-null   int64
 23  T+12_month_activity     8448 non-null   int64
 24  Transactor_revolver     8448 non-null   object
 25  avg_spends_l3m          8448 non-null   int64
 26  Occupation_at_source    8448 non-null   object
 27  cc_limit                8448 non-null   int64
dtypes: datetime64[ns](1), int64(19), object(8)
memory usage: 1.8+ MB
```

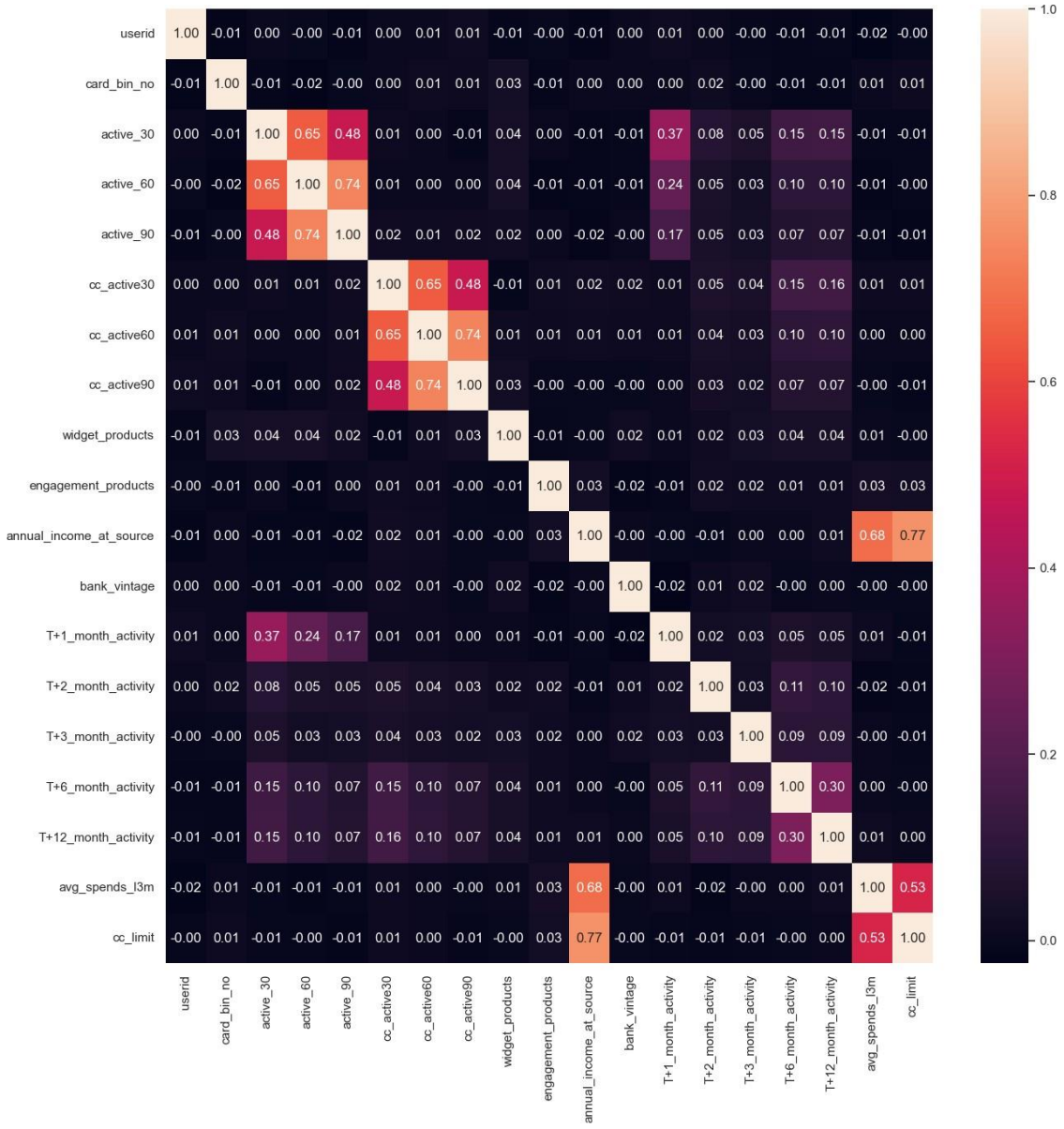
Number of duplicate rows = 0

	avg_spends_l3m	cc_limit
count	8448.000000	8448.000000
mean	49527.365530	251706.912879
std	46244.954836	229114.856385
min	0.000000	0.000000
25%	17110.000000	90000.000000
50%	37943.000000	150000.000000
75%	66095.750000	350000.000000
max	289292.000000	990000.000000




4000
3500
3000
2500
2000
1500
1000
500
0
count
Self Employed
0
Student
Salaried
Retired
Housewife
Occupation_at_source


3500
3000
2500
2000
1500
1000
500
0
count
Occupation_at_source
Self Employed
0
Student
Salaried
Retired
Housewife
Visa
Mastercard
Amex
Issuer


userid 1.00 -0.01 0.00 -0.00 -0.01 0.00 0.01 0.01 -0.01 -0.00 -0.01 0.00 0.01 0.00 -0.00 -0.01 -0.01 -0.02 -0.00
card_bin_no -0.01 1.00 -0.01 -0.02 -0.00 0.00 0.01 0.01 0.03 -0.01 0.00 0.00 0.00 0.02 -0.00 -0.01 -0.01 0.01 0.01
active_30 0.00 -0.01 1.00 0.65 0.48 0.01 0.00 -0.01 0.04 0.00 -0.01 -0.01 0.37 0.08 0.05 0.15 0.15 -0.01 -0.01
active_60 -0.00 -0.02 0.65 1.00 0.74 0.01 0.00 0.00 0.04 -0.01 -0.01 -0.01 0.24 0.05 0.03 0.10 0.10 -0.01 -0.00
active_90 -0.01 -0.00 0.48 0.74 1.00 0.02 0.01 0.02 0.02 0.00 -0.02 -0.00 0.17 0.05 0.03 0.07 0.07 -0.01 -0.01
cc_active30 0.00 0.00 0.01 0.01 0.02 1.00 0.65 0.48 -0.01 0.01 0.02 0.02 0.01 0.05 0.04 0.15 0.16 0.01 0.01
cc_active60 0.01 0.01 0.00 0.00 0.01 0.65 1.00 0.74 0.01 0.01 0.01 0.01 0.01 0.04 0.03 0.10 0.10 0.00 0.00
cc_active90 0.01 0.01 -0.01 0.00 0.02 0.48 0.74 1.00 0.03 -0.00 -0.00 -0.00 0.00 0.03 0.02 0.07 0.07 -0.00 -0.01
widget_products -0.01 0.03 0.04 0.04 0.02 -0.01 0.01 0.03 1.00 -0.01 -0.00 0.02 0.01 0.02 0.03 0.04 0.04 0.01 -0.00
engagement_products -0.00 -0.01 0.00 -0.01 0.00 0.01 0.01 -0.00 -0.01 1.00 0.03 -0.02 -0.01 0.02 0.02 0.01 0.01 0.03 0.03
annual_income_at_source -0.01 0.00 -0.01 -0.01 -0.02 0.02 0.01 -0.00 -0.00 0.03 1.00 -0.00 -0.00 -0.01 0.00 0.00 0.01 0.68 0.77
bank_vintage 0.00 0.00 -0.01 -0.01 -0.00 0.02 0.01 -0.00 0.02 -0.02 -0.00 1.00 -0.02 0.01 0.02 -0.00 0.00 -0.00 -0.00
T+1_month_activity 0.01 0.00 0.37 0.24 0.17 0.01 0.01 0.00 0.01 -0.01 -0.00 -0.02 1.00 0.02 0.03 0.05 0.05 0.01 -0.01
T+2_month_activity 0.00 0.02 0.08 0.05 0.05 0.05 0.04 0.03 0.02 0.02 -0.01 0.01 0.02 1.00 0.03 0.11 0.10 -0.02 -0.01
T+3_month_activity -0.00 -0.00 0.05 0.03 0.03 0.04 0.03 0.02 0.03 0.02 0.00 0.02 0.03 0.03 1.00 0.09 0.09 -0.00 -0.01
T+6_month_activity -0.01 -0.01 0.15 0.10 0.07 0.15 0.10 0.07 0.04 0.01 0.00 -0.00 0.05 0.11 0.09 1.00 0.30 0.00 -0.00
T+12_month_activity -0.01 -0.01 0.15 0.10 0.07 0.16 0.10 0.07 0.04 0.01 0.01 0.00 0.05 0.10 0.09 0.30 1.00 0.01 0.00
avg_spends_l3m -0.02 0.01 -0.01 -0.01 -0.01 0.01 0.00 -0.00 0.01 0.03 0.68 -0.00 0.01 -0.02 -0.00 0.00 0.01 1.00 0.53
cc_limit -0.00 0.01 -0.01 -0.00 -0.01 0.01 0.00 -0.01 -0.00 0.03 0.77 -0.00 -0.01 -0.01 -0.01 -0.00 0.00 0.53 1.00
userid
card_bin_no
active_30
active_60
active_90
cc_active30
cc_active60
cc_active90
widget_products
engagement_products
annual_income_at_source
bank_vintage
T+1_month_activity
T+2_month_activity
T+3_month_activity
T+6_month_activity
T+12_month_activity
avg_spends_l3m
cc_limit
1.0
0.8
0.6
0.4
0.2
0.0

	card_no	card_bin_no	Issuer	card_type	high_networth	active_30	active_60	active_90
0	438439XXXX	438439	Visa	edge	B	0	1	1
1	437748XXXX	437748	Visa	prosperity	A	1	1	1
2	437748XXXX	437748	Visa	rewards	C	0	0	0
3	425806XXXX	425806	Visa	indianoil	E	0	1	1
4	437748XXXX	437748	Visa	edge	B	1	1	1

	card_bin_no	active_30	active_60	active_90	cc_active30	cc_active60	cc_ac
count	8448.000000	8448.000000	8448.000000	8448.000000	8448.000000	8448.000000	8448.0
mean	0.406290	0.292377	0.494792	0.642045	0.284091	0.484493	0.6
std	0.207044	0.454881	0.500002	0.479427	0.451007	0.499789	0.4
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.0
25%	0.334947	0.000000	0.000000	0.000000	0.000000	0.000000	0.0
50%	0.411749	0.000000	0.000000	1.000000	0.000000	0.000000	1.0
75%	0.417779	1.000000	1.000000	1.000000	1.000000	1.000000	1.0
max	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.0

	card_no	Issuer	card_type	high_networth	hotlist_flag	other_bank_cc_holding	Transa
0	4384 39XX XXXX	Visa	edge	B	N		Y
1	4377 48XX XXXX	Visa	prosperity	A	N		Y
2	4377 48XX XXXX	Visa	rewards	C	N		N
3	4258 06XX XXXX	Visa	indianoil	E	N		N
4	4377 48XX XXXX	Visa	edge	B	N		N
...	...	...	...	...	...		...
8443	4262 41XX XXXX 37691	Visa	chartered	A	N		N

card_no	object	card_bin_no	int64
Issuer	category		
card_type	object		
high_networth	object		
active_30	int64		
active_60	int64		
active_90	int64		
cc_active30	int64		
cc_active60	int64		
cc_active90	int64		
hotlist_flag	object		
widget_products	int64		
engagement_products	int64		
annual_income_at_source	int64		
other_bank_cc_holding	object		
bank_vintage	int64		

T+1_month_activity	int64
T+2_month_activity	int64
T+3_month_activity	int64
T+6_month_activity	int64
T+12_month_activity	int64
Transactor_revolver	object
avg_spends_l3m	int64
Occupation_at_source	object
cc_limit	int64

dtype: object

Insights: Changing the label doesnot effect the card details of average spending.